

Functional Hyperedges: Connecting Embryonic Development to Future Neural Circuitry

Functional Hyperedges and Feed-Forward Loop (FFL) Motifs

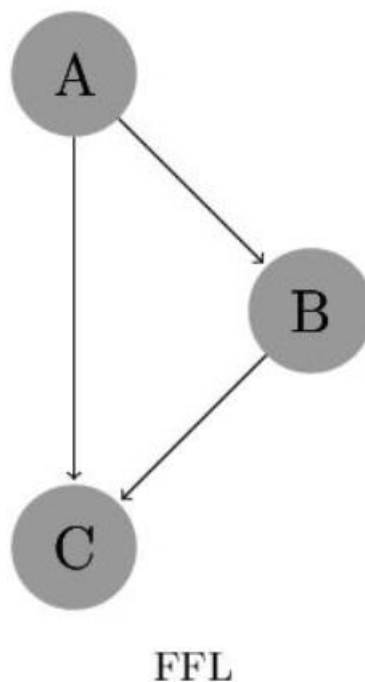
A Functional Hyperedge is a higher-order relationship connecting embryonic cells whose descendant neurons participate in a common Feed-Forward Loop (FFL) motif in the adult *C. elegans* connectome.

A Feed-Forward Loop is a three-neuron motif with the following structure:

$A \rightarrow B$

$A \rightarrow C$

$B \rightarrow C$



Unlike a standard graph edge, which connects only two nodes, a Functional Hyperedge represents the entire three-neuron motif as a single biological unit. Each FFL therefore becomes one hyperedge connecting the three participating cells

Why is this Important in Developmental Biology?

Developmental datasets typically describe:

- Cell lineage
- Cell position
- Cell movement
- Temporal organization

However, they do not explicitly capture how cells will eventually contribute to neural circuits.

Two embryonic cells may originate from different lineage branches and remain spatially distant throughout development, yet later become part of the same functional circuit in the mature nervous system.

Functional Hyperedges introduce this missing layer of information by linking embryonic cells according to the neural motifs they will eventually form.

Impact of Using Functional Hyperedges

Functional Hyperedges provide a biologically meaningful prior derived from the adult connectome.

By incorporating future circuit membership into the developmental hypergraph, the model gains access to information that is not available through spatial, lineage, or temporal relationships alone.

Potential benefits include:

- Improved representation of long-range developmental relationships.
 - Identification of cells that contribute to common neural circuits despite developmental separation.
 - Enhanced biological interpretability of learned embeddings.
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Functional Lookup Table

The Functional Lookup Table serves as a bridge between embryonic development and the adult connectome.

The table is constructed in four stages:

1. Map adult neurons to their embryonic lineage identities.
2. Extract Feed-Forward Loop motifs from the adult connectome.
3. Convert each neuronal triplet into the corresponding embryonic cell triplet.
4. Assign a weight based on the total synaptic connectivity of the motif.

Table overview

Motif ID	Adult A (FFL source)	Adult B (intermediary)	Adult C (FFL target)	Embryo A Sulston normalized	Embryo B Sulston normalized	Embryo C Sulston normalized	Syn A→B	Syn A→C	Syn B→C	Total Synapses	w_e =log(1+total)
FFL_0000	ADAL	AIBL	AVBL	Abplapaaaapp	ABplaapappa	ABplpaapaap	1	4	5	10	2.397895
FFL_0001	ADAL	AIBL	RIML	ABplapaaaapp	ABplaapappa	ABplppaapap	1	3	2	6	1.94591
FFL_0002	ADAL	AIBR	AVAR	ABplapaaaapp	ABpraapappa	ABalaappapa	2	2	1	5	1.791759
FFL_0003	ADAL	AIBR	AVBR	ABplapaaaapp	ABpraapappa	ABprpaapaap	2	7	3	12	2.564949

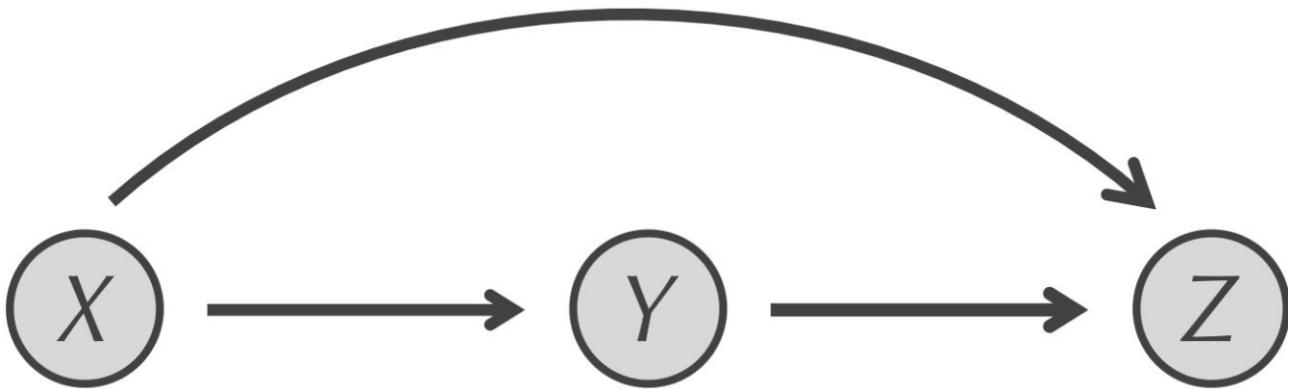
Each row represents one Feed-Forward Loop motif and one Functional Hyperedge.

The adult neuron columns identify the three neurons that participate in a Feed-Forward Loop motif within the mature connectome. These neurons are then mapped to their embryonic lineage identities, and the resulting embryonic cell triplet forms the Functional Hyperedge represented in the developmental hypergraph.

The weight is calculated from the total synaptic connectivity of the motif using:

$$w = \log(1 + \text{Total Synapses})$$

which reduces the influence of extremely large synapse counts while preserving relative differences between motifs.



Key Insight

The Functional Lookup Table answers a simple but important question:

Which embryonic cells will later participate in the same functional neural motif?

By answering this question for every Feed-Forward Loop in the connectome, the table creates a direct connection between embryogenesis and mature neural circuitry, enabling future functional organization to be incorporated into the Dynamic Hypergraph framework.